Invasive Plant Survey - Belt Transect Protocol

This protocol will be followed for surveying invasive plants near trails in Reno, Nevada. Each student group will survey the presence and absence of species invasive to the Great Basin at a sampling site location using the *belt transect* survey method. At each sampling site location, 2 survey collection points will be selected along a trail by the student group, which will serve as the starting points for each of the 2 *belt transects*. Belt transects will be 30 meters in length and 4 meters in width (2 meters to the right of the trail and 2 meters to the left of the trail), divided into 1 meter by 1 meter sampling squares (see *figure 1*). Students will use a 30m tape measure to measure the 30 meter length of the transect and a tape measure to measure the width. Students will be provided an invasive species guide to help identify plant species that are present in each sampling square, and a data sheet will be used to record the data.

1. From a list provided, students will select a sampling site location within Reno, NV, where they will conduct 2 belt transect surveys.
2. Choose a survey collection points on the trail, within the boundaries of the sampling site location.
3. Measure 30 meters from the survey collection point parallel to the trail edge using the 30m tape measure. Even as the trail bends or terrain is uneven, the transect line should remain on the trail and parallel to the trail edge.
4. Flag (or mark) the start and end points of your transect line clearly. Record the start and end GPS coordinates for each belt transect on the data sheet or in the web app.
5. Record the start and end times for surveying each belt transect.
6. For photo documentation: Take four photos for each transect.
    a. Take a photo (landscape, 1x zoom) from the 0m mark of the transect line looking up the transect line.
    b. Take a photo (landscape, 1x zoom) from the 30m mark of the transect line looking down the transect line.
    c. Take a photo (landscape, 1x zoom) from the 15m mark of the transect line looking left
    d. Take a photo (landscape, 1x zoom) from the 15m mark of the transect line looking right
7. Starting at the 0-1 meter mark of the transect line, measure (using a tape measure) 2 meters to the *left* (from the left trail edge) perpendicular to the transect line. This establishes your 2 x 1^2 meter sampling squares on the left side for the 0-1m segment. Continue this process for every meter of your transect line (1-2m, 2-3m…29-30m). For every 1 meter of the transect line that is surveyed, 2 x 1^2 meter sampling squares will be surveyed to the left of the left trail edge.
8. Follow the same procedure for the *right* side of your belt transect. Starting at the 0-1 meter mark of the transect line, measure (using a tape measure) 2 meters to the *right*

(from the right trail edge) perpendicular to the transect line. This establishes your 2 x 1^2 meter sampling squares on the right side for the 0-1m segment. Continue this process for every meter of your transect line (1-2m, 2-3m…29-30m). For every 1 meter of the transect line that is surveyed, 2 x 1^2 meter sampling squares will be surveyed to the right of the right trail edge. This is clearly represented in *figure 1*.

9. Within each 1 x 1 meter sampling square, record on the data sheet which species (using the invasive species guide) are present. A species is considered present if any portion of its aerial canopy is within the sampling square. Record all species that canopy overlaps the sampling square, including understory and groundcover species.
10. In the correlating cell, on the data sheet or web app, record the plant species code for species present within a sampling square (see species list), NS (for not surveyed), or a blank cell (for unrecorded).
11. Repeat steps 2-10 for belt transects two and three.
12. Input plant data and photo documentation into the web app. If recording on a physical data sheet, transcribe the data from the data sheet to the web app after the survey is complete.

Definitions

- *Belt transect*: A monitoring technique used to measure frequency, and presence/absence of plant or animal species along a transect line using quadrats.
- *Left*: The left side of your transect will be determined by facing from the 0m mark towards the 30m mark.
- *Right*: The right side of your transect will be determined by facing from the 0m mark towards the 30m mark.

1-2m	0-1m	Trail	0-1m	1-2m

Figure 1: This table demonstrates the layout of the belt transect. The 30 meter transect line will be measured on the trail. The width of the transect belt will be measured 2 meters to the right of the right trail edge and 2 meters to the left of the left trail edge perpendicular to the trail. Each sampling square will be 1 meter by 1 meter.